

seminar note

A harmonic view of random walks and electrical networks

Goal: explain the connection, then use Rayleigh shorting to show $\mathbb{Z}^2$ is recurrent

Main spine.

harmonicity $\implies$ same Dirichlet problem $\implies$ uniqueness $\implies$ RW $\leftrightarrow$ network dictionary $\implies R_{\text{eff}}(0 \leftrightarrow \infty)$

0. Opening (about 40–60 seconds)

What I want to say.

Today I want to explain one conceptual point:

Random walk problems become electrical network problems because both sides are governed by the same notion of *harmonicity*.

Then I want to show why this reformulation is useful: Rayleigh monotonicity turns recurrence into a resistance estimate, and in dimension 2 the estimate is easy after shorting square shells.

One-sentence destination.

$$\mathbb{Z}^2 \text{ is recurrent because } R_{\text{eff}}(0 \leftrightarrow \infty) = \infty.$$

1. Harmonicity is the common language (about 1.5 min)

Fix a finite graph $G = (V, E)$ for the moment.

If x is an interior vertex, call f *harmonic at x* if

$$f(x) = \frac{1}{\deg(x)} \sum_{y \sim x} f(y).$$

So the value at a point is the average of the neighboring values.

Line to say aloud. The key point is that the common structure is not “randomness” and not “electricity”; it is this local averaging law.

Optional remark if needed. For weighted networks the averaging uses conductances, but for the short talk I only need the unit-conductance/simple-random-walk case.

2. Probability side: hitting probabilities are harmonic (about 2 min)

Let A, B be two disjoint boundary sets in a finite graph, and define

$$h(x) = \mathbb{P}_x(\tau_A < \tau_B).$$

Boundary values:

$$h|_A = 1, \quad h|_B = 0.$$

At an interior vertex x ,

$$h(x) = \frac{1}{\deg(x)} \sum_{y \sim x} h(y).$$

Reason. First-step decomposition: from x the walk first jumps to one of its neighbors, and then the same question restarts there.

What to emphasize.

- This is the probabilistic source of harmonicity.
- It is exactly the discrete mean-value property.
- I do *not* need an explicit formula for h .

Sentence to say aloud. On the random-walk side, harmonicity comes from conditioning on the first step.

3. Electrical side: voltages are harmonic (about 2 min)

Now put unit resistors on the same graph.

Prescribe boundary voltages

$$v|_A = 1, \quad v|_B = 0.$$

At an interior node x , Kirchhoff's current law says net current is zero:

$$\sum_{y \sim x} (v(x) - v(y)) = 0.$$

Hence

$$v(x) = \frac{1}{\deg(x)} \sum_{y \sim x} v(y).$$

So voltage satisfies the *same* harmonic equation with the *same* boundary values.

Line to say aloud.

Hitting probabilities and voltages solve the same Dirichlet problem.

Important tactical note to myself. Do not spend time on general conductance notation here. The point is the structural identity, not maximal generality.

4. Uniqueness gives the dictionary (about 2 min)

Principle. If two harmonic functions agree on the boundary, then they agree everywhere.

Proof idea: if u, v are harmonic and $u = v$ on the boundary, then $w = u - v$ is harmonic and vanishes on the boundary. By the maximum principle, $w \equiv 0$.

Therefore,

$$h(x) = v(x).$$

This is the real conceptual payoff.

Harmonicity gives resemblance. Uniqueness upgrades resemblance into identification.

Dictionary box.

hitting probability $\longleftrightarrow$ voltage

If I need one more sentence. This is why electrical methods are not just an analogy; they actually compute probabilistic quantities.

5. Why the connection is useful: recurrence becomes resistance (about 2 min)

Now move to the infinite lattice $\mathbb{Z}^d$ with unit resistors on all edges.

Let $G(r)$ be a finite exhaustion around the origin, e.g. a box or ball, and let $\partial G(r)$ be its outer boundary.

Define $R_{\text{eff}}^{(r)}$ as the effective resistance from 0 to $\partial G(r)$.

For the finite wired problem,

$$p_{\text{esc}}^{(r)} = \frac{1}{2d R_{\text{eff}}^{(r)}}.$$

Passing to the limit,

$$R_{\text{eff}} := \lim_{r \rightarrow \infty} R_{\text{eff}}^{(r)}, \quad p_{\text{esc}} = \frac{1}{2d R_{\text{eff}}}.$$

Hence

$$p_{\text{esc}} = 0 \iff R_{\text{eff}} = \infty.$$

So

random walk is recurrent $\iff R_{\text{eff}}(0 \leftrightarrow \infty) = \infty$.

What to stress aloud.

- This is where the electrical reformulation becomes useful.
- Recurrence is now an estimate of a deterministic network quantity.

Sentence to say aloud.

A probabilistic return question has become an electrical resistance estimate.

6. Rayleigh monotonicity (about 1 min)

Two basic operations:

- **Shorting law:** identifying vertices can only *decrease* effective resistance.
- **Cutting law:** deleting edges can only *increase* effective resistance.

Symbolically,

$$R_{\text{eff}}(\text{shorted}) \leq R_{\text{eff}}(\text{original}) \leq R_{\text{eff}}(\text{cut}).$$

What I need here. Only the shorting law.

Sentence to say aloud.

Once the problem is rewritten in terms of resistance, Rayleigh gives us a way to estimate it by simplifying the network.

7. Application: $\mathbb{Z}^2$ is recurrent (about 4 min)

This is the key proof of the talk.

Step 1: square shells

Let

$$Q_n = [-n, n]^2 \cap \mathbb{Z}^2, \quad C_n = \partial Q_n.$$

Think of C_n as the n -th square shell around the origin.

What to say. We decompose the lattice into concentric square shells.

Step 2: short each shell to one node

Short together all vertices belonging to the same shell C_n .

Then the network collapses to a one-dimensional chain

$$0 \longleftrightarrow 1 \longleftrightarrow 2 \longleftrightarrow 3 \longleftrightarrow \dots$$

where the edge between n and $n + 1$ is actually a bundle of parallel edges.

By the shorting law,

$$R_{\text{eff}}(\text{shorted network}) \leq R_{\text{eff}}(\mathbb{Z}^2).$$

So if the shorted network already has infinite resistance, then so does the original lattice.

Step 3: count branches between shells

Between shell C_n and shell C_{n+1} there are

$$8n + 4$$

unit-resistance edges.

Check small cases:

$$n = 0 : 4, \quad n = 1 : 12, \quad n = 2 : 20.$$

What to say. This is exactly what appears in the figure after shorting: 4 branches, then 12 branches, then 20 branches, and in general $8n + 4$.

Step 4: replace each bundle by one resistor

A bundle of $8n + 4$ parallel unit resistors is equivalent to a single resistor of value

$$\frac{1}{8n + 4}.$$

So the shorted network becomes a series network:

$$\frac{1}{4}, \quad \frac{1}{12}, \quad \frac{1}{20}, \quad \dots, \quad \frac{1}{8n + 4}, \quad \dots$$

Step 5: series sum diverges

Therefore

$$R_{\text{eff}}(\text{shorted network}) = \sum_{n=0}^{\infty} \frac{1}{8n + 4} = \infty.$$

Since shorting can only decrease resistance,

$$R_{\text{eff}}(\mathbb{Z}^2) \geq R_{\text{eff}}(\text{shorted network}) = \infty.$$

Hence

$$R_{\text{eff}}(0 \leftrightarrow \infty) = \infty.$$

By the recurrence/resistance criterion,

$\text{simple random walk on } \mathbb{Z}^2 \text{ is recurrent.}$

Ideal spoken ending.

So the proof is not a delicate probabilistic argument. The harmonic viewpoint lets us translate recurrence into a resistance question, and then Rayleigh shorting makes the resistance estimate almost visual.

8. Last 20 seconds / closing line

If I have time, I can end with:

Dimension 3 goes the other way: one tries to find a residual subnetwork of finite resistance, typically via cutting rather than shorting.

But I should *not* try to explain the $d = 3$ proof in this short talk.

Very short blackboard-style summary for myself

1. harmonic = local averaging.
 2. hitting probability is harmonic.
 3. voltage is harmonic.
 4. same boundary values + uniqueness $\Rightarrow$ same function.
 5. recurrence $\iff R_{\text{eff}}(0 \leftrightarrow \infty) = \infty$.
 6. Rayleigh shorting decreases resistance.
 7. short square shells in $\mathbb{Z}^2$.
 8. get $8n + 4$ parallel edges between n and $n + 1$.
 9. equivalent series network has resistors $\frac{1}{8n+4}$.
 10. $\sum_{n \geq 0} \frac{1}{8n+4} = \infty$.
 11. therefore $\mathbb{Z}^2$ recurrent.
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Warnings to myself

- Do not get lost in general conductances.
- Do not prove the maximum principle in detail.
- Do not spend too long justifying $8n + 4$; let the picture do the work.
- Keep repeating the conceptual chain:

harmonic $\rightarrow$ Dirichlet problem $\rightarrow$ uniqueness $\rightarrow$ resistance $\rightarrow$ Rayleigh.

- The talk is about the *connection*, not about listing many theorems.